COLIMIT DESCRIPTION OF F_! OVER A DIRECTED OPEN COVERING

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Allowed standard prerequisites (editorial).

- Covariance of f_! for separated morphisms (Remark remark-covariance-f-shriek-separated, canonical tag 0F53). cited at statement lines 935-936. Supplies the transition maps of the colimit.
- Sections of directed colimits (Sites, Lemma sites-lemma-directed-colimits-sections, canonical tag 0738). cited at proof lines 945-947. Evaluates the directed colimit on a quasi-compact object.

Source reference. Original remark 0F53: sources/stacks-chow/tags.txt, source lines 18923–18923. Cited at statement lines 935-936: supplies the transition maps of the colimit.

Source reference. Original lemma 0738: sources/stacks-chow/tags.txt, source lines 9052–9052. Cited at proof lines 945-947: evaluates the directed colimit on a quasi-compact object.

Lemma 0.2. *Let $f : X \to Y$ be morphism of schemes which is separated and locally of finite type. Let $X = \bigcup_{i \in I} X_i$ be an open covering such that for all $i, j \in I$ there exists a k with $X_i \cup X_j \subset X_k$. Denote $f_i : X_i \to Y$ the restriction of f. Then*

$$f_! \mathcal{F} = \operatorname{colim}_{i \in I} f_{i,!}(\mathcal{F}|_{X_i})$$

functorially in $\mathcal{F} \in Ab(X_{étale})$ where the transition maps are the ones constructed in Remark 0F53.

Proof. It suffices to show that the canonical map from right to left is a bijection when evaluated on a quasi-compact object V of $Y_{étale}$. Observe that the colimit on the right hand side is directed and has injective transition maps. Thus we can use Sites, Lemma 0738 to evaluate the colimit. Hence, the statement comes down to the observation that a closed subset $Z \subset X_V$ proper over V is quasi-compact and hence is contained in $X_{i,V}$ for some i. □

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